

AI Agent Guide: Automate Data Board

Action name: auto_data_board

What it does: Performs CRUD operations on Data Board records. One action with 7 sub-operations selected via `chooseAction`.

❑ CRITICAL: DYNAMIC PROPERTY NESTING RULE

The platform treats `inputMethod`, `contentJson`, `additionalFields`, `identifier`, `identifierNested`, `recordId`, and `outputData` as **DYNAMIC** properties. Their values **MUST** be wrapped inside an object keyed by the property name itself. Flat values will be **silently nullified** by the platform.

Wrong (flat — will be nullified):

```
"inputMethod": "json",
"contentJson": {"<field_id>": "value"},
"identifier": "fieldsOnBoard",
"outputData": "allData"
```

Correct (nested — required):

```
"inputMethod": {"inputMethod": "json"},
"contentJson": {"contentJson": {"<field_id>": "value"}},
"identifier": {"identifier": "fieldsOnBoard"},
"outputData": {"outputData": "allData"},
"recordId": {"recordId": "<record_id>"}
```

`propertySettings` for DYNAMIC properties **MUST** include a `schema` block. See the FULL CONFIGURATION EXAMPLES in each action section for the correct format.

This nesting rule applies to **ALL** configuration examples below.

❑ CRITICAL: FIELD IDS — NOT FIELD NAMES

This is the **#1 runtime failure cause**. The board API rejects requests with field names.

RULE: Keys in `contentJson` and values in `additionalFields.actionPairs[].field` **MUST** be MongoDB ObjectId-style `field_ids` (24-character hex strings like `"69ca065d5817f89ead6fadf4"`).

NEVER use display names like `"Email"`, `"Name"`, `"Phone"`, `"full_name"`, `"email"`, `"address"`, etc. These will cause: `{"message": "Identifier value is required"}` at runtime.

Where to get field IDs:

- From the parent orchestrator/task prompt** — if the orchestrator passes board details with `field_id` values, use those EXACT IDs.
- From `boardDetails` action** — add a step with `chooseAction: "boardDetails"` to retrieve the full board schema including all `field_ids`.
- From the Databoard Agent result** — if a Databoard Agent created the board, its result contains field IDs. Use them directly.

WRONG (will fail at runtime):

```
"contentJson": {"contentJson": {
  "name": "{{step_1['success'] [0] ['data'] ['name']}}",
  "email": "{{step_1['success'] [0] ['data'] ['email']}}"
}}
```

CORRECT (uses 24-char hex field IDs):

```
"contentJson": {"contentJson": {
  "69ca065d5817f89ead6fadf4": "{{step_1['success'] [0] ['data'] ['name']}}",
  "69ca065d5817f89ead6fadf5": "{{step_1['success'] [0] ['data'] ['email']}}"
}}
```

❑ CRITICAL: DOCUMENT AI OUTPUT PATH

When wiring Document AI output to Data Board fields, the extracted data is **DIRECTLY** inside `data`. There is **NO** `properties` wrapper.

```
WRONG:  {{step_1['name']}}           ← missing success[0].data path
WRONG:  {{step_1['email']}}          ← missing success[0].data path
WRONG:  {{step_1['success'][0]['data']['properties']['full_name']}} ← properties does NOT exist
CORRECT: {{step_1['success'][0]['data']['full_name']}} ← direct data access
CORRECT: {{step_1['success'][0]['data']['email']}} ← direct data access
```

Phone fields return an object. Use a string sub-key:

```
{{step_1['success'][0]['data']['phone']['calling_code_with_number']}}
```

CRITICAL CONFIGURATION WARNINGS — READ FIRST

- inputMethod IS MANDATORY FOR create AND update ACTIONS.** Must use nested format: `{"inputMethod": "json"}` or `{"inputMethod": "field"}`. If missing, step WILL fail validation.
- ALL DYNAMIC PROPERTIES MUST USE NESTED FORMAT.** Every DYNAMIC property set in `input` MUST use the nested wrapping format shown above.
- ALL DYNAMIC PROPERTIES MUST HAVE schema IN propertySettings.** Missing schema entries will cause silent validation failures.
- CONDITIONAL FIELDS MUST MATCH inputMethod AND identifier VALUES.**
 - If `inputMethod = "json"` → you MUST set `contentJson` (nested)
 - If `inputMethod = "field"` → you MUST set `additionalFields` with `actionPairs`
 - If `identifier = "recordId"` → you MUST set `recordId` (nested)
 - If `identifier = "fieldsOnBoard"` → you MUST set `identifierNested`
- FIELD IDS, NOT FIELD NAMES** (see dedicated section above).
- version AND chooseAction ARE STATIC_DROPDOWN — MUST INCLUDE schema IN propertySettings.**
- IDENTIFIER FIELD MUST HAVE A NON-EMPTY VALUE.** Every board has one field marked `is_identifier: true`. When creating a record, this field MUST have a non-empty value. Use `{{trigger.body['file_url']}}` or another trigger value — NOT an OCR field that may be empty.
- ALL VALUES IN contentJson MUST BE STRINGS.** If an upstream step returns an object, reference a specific string sub-field.
 - WRONG: `"<field_id>": "{{step_1['success'][0]['data']}}"` — passes `[object Object]`
 - CORRECT: `"<field_id>": "{{step_1['success'][0]['data']['full_name']}}"` — passes a string
- DO NOT USE HANDLEBARS {{#if}} CONDITIONALS IN FIELD VALUES.** Use a Code step for conditional logic instead.

1. Common Settings (Always Visible)

Property	Type	Required	Default	Resolution Method
workflowName	ShortText	No	"Automate Data Board"	Direct value
version	StaticDropdown	Yes	"1"	Read from <code>options.options</code> in <code>get_piece_details</code>
boardId	Dropdown	Yes	—	Call <code>get_piece_options</code>
chooseAction	StaticDropdown	Yes	—	Read from <code>options.options</code> in <code>get_piece_details</code>

2. Actions and Their Settings

ACTION: `create` — Create a Record

Property	Type	Required	Description
inputMethod	StaticDropdown	YES — MANDATORY	"field" or "json"

If `inputMethod = "json"`:

Property	Type Required	Description
contentJson	JSON Yes	JSON object mapping field IDs to values.

If inputMethod = "field":

Property	Type Required	Description
additionalFields.actionPairs	Array No	Array of { field: "<field_id>", value: "<value>" } pairs.

❑ CRITICAL: inputMethod VALID VALUES

inputMethod is a DYNAMIC property. It MUST use nested format and ONLY these two values:

- {"inputMethod": "json"} — use with contentJson
- {"inputMethod": "field"} — use with additionalFields.actionPairs

NEVER use any other value. The following are WRONG and will cause silent failures:

```

WRONG: {"mode": "useAdditionalFields"}
WRONG: {"inputMethod": "useAdditionalFields"}
WRONG: "json" (flat, not nested)
WRONG: "field" (flat, not nested)

```

❑ CRITICAL: propertySettings for STATIC_DROPDOWN — NO schema block

STATIC_DROPDOWN properties (version, chooseAction) **MUST** have propertySettings as {"type": "MANUAL"} with NO schema block. Adding a schema block causes AP engine crash: PropertyTypeNotExecutableError: Property type is not executable: STATIC_DROPDOWN.

```

WRONG: "version": {"type": "MANUAL", "schema": {"version": {"type": "STATIC_DROPDOWN", ...}}}
CORRECT: "version": {"type": "MANUAL"}

```

Only DYNAMIC properties (inputMethod, contentJson, additionalFields, identifier, etc.) need schema blocks.

❑ CRITICAL: errorHandlingOptions PLACEMENT

errorHandlingOptions belongs ONLY at settings.errorHandlingOptions. **NEVER** put errorHandlingOptions inside propertySettings — this causes AP validation error: "propertySettings/errorHandlingOptions must have required property 'type'".

```

WRONG: settings.propertySettings.errorHandlingOptions = { "continueOnFailure": ... }
CORRECT: settings.errorHandlingOptions = { "continueOnFailure": { "value": false }, "retryOnFailure": { "value": false } }

```

FULL CONFIGURATION EXAMPLE — create with inputMethod = "json"

```

{
  "name": "step_2",
  "displayName": "Create Databoard Record",
  "type": "PIECE",
  "valid": true,
  "skip": false,
  "settings": {
    "pieceName": "@activepieces/piece-automate-data-board",
    "pieceVersion": "~0.0.1",
    "actionName": "auto_data_board",
    "input": {
      "workflowName": "Automate Data Board",
      "version": "1",
      "boardId": "<resolved_board_id_from_get_piece_options>",
      "chooseAction": "create",
      "inputMethod": {"inputMethod": "json"},
      "contentJson": {"contentJson": {
        "<24_char_hex_field_id_1>": "{{step_1['success'][0]['data']['full_name']}}",
        "<24_char_hex_field_id_2>": "{{step_1['success'][0]['data']['email']}}",
        "<24_char_hex_field_id_3>": "{{step_1['success'][0]['data']['phone']['calling_code_with_number']}}",
        "<identifier_field_id>": "{{trigger.body['file_url']}}"
      }},
      "additionalFields": {},
      "identifier": {},
      "identifierNested": {},
      "recordId": {},
      "outputData": {},
      "testing_organization_id": "<org_id>"
    },
    "propertySettings": {
      "workflowName": { "type": "MANUAL" },
      "version": { "type": "MANUAL" },
      "boardId": { "type": "MANUAL" },
      "chooseAction": { "type": "MANUAL" },
      "inputMethod": {
        "type": "MANUAL",
        "schema": {
          "inputMethod": {
            "type": "STATIC_DROPDOWN",
            "options": {"options": [{"label": "Fields", "value": "field"}, {"label": "JSON", "value": "json"}]},
            "required": true,
            "displayName": "Input Method"
          }
        }
      },
      "contentJson": {
        "type": "MANUAL",
        "schema": {
          "contentJson": {"type": "JSON", "required": true, "displayName": "Content (JSON)"}
        }
      },
      "additionalFields": { "type": "MANUAL", "schema": {} },
      "identifier": { "type": "MANUAL", "schema": {} },
      "identifierNested": { "type": "MANUAL", "schema": {} },
      "recordId": { "type": "MANUAL", "schema": {} },
      "outputData": { "type": "MANUAL", "schema": {} },
      "testing_organization_id": {
        "type": "MANUAL",
        "schema": {
          "testing_organization_id": {"type": "SHORT_TEXT", "required": false, "displayName": "Organization ID"}
        }
      }
    }
  },
}

```

```
"errorHandlingOptions": {  
  "continueOnFailure": { "value": false },  
  "retryOnFailure": { "value": false }  
}  
}  
}
```

FULL CONFIGURATION EXAMPLE — create with inputMethod = "field"

- When using inputMethod: "field", data goes in additionalFields.actionPairs as an array of {field, value} pairs. The field MUST be a 24-char hex field ID, NOT a field name.

```

{
  "name": "step_2",
  "displayName": "Create Databoard Record",
  "type": "PIECE",
  "valid": true,
  "skip": false,
  "settings": {
    "pieceName": "@activepieces/piece-automate-data-board",
    "pieceVersion": "~0.0.1",
    "actionName": "auto_data_board",
    "input": {
      "workflowName": "Automate Data Board",
      "version": "1",
      "boardId": "<resolved_board_id_from_get_piece_options>",
      "chooseAction": "create",
      "inputMethod": {"inputMethod": "field"},
      "additionalFields": {"actionPairs": [
        {"field": "<24_char_hex_identifier_field_id>", "value": "{{trigger.body['file_url']}}"},
        {"field": "<24_char_hex_field_id_1>", "value": "{{step_1['success'][0]['data']['full_name']}}"},
        {"field": "<24_char_hex_field_id_2>", "value": "{{step_1['success'][0]['data']['email']}}"},
        {"field": "<24_char_hex_field_id_3>", "value": "{{step_1['success'][0]['data']['phone']['calling_code_with_number']}}"}
      ]},
      "contentJson": {},
      "identifier": {},
      "identifierNested": {},
      "recordId": {},
      "outputData": {},
      "testing_organization_id": "<org_id>"
    },
    "propertySettings": {
      "workflowName": { "type": "MANUAL" },
      "version": { "type": "MANUAL" },
      "boardId": { "type": "MANUAL" },
      "chooseAction": { "type": "MANUAL" },
      "inputMethod": {
        "type": "MANUAL",
        "schema": {
          "inputMethod": {
            "type": "STATIC_DROPDOWN",
            "options": {"options": [{"label": "Fields", "value": "field"}, {"label": "JSON", "value": "json"}]},
            "required": true,
            "displayName": "Input Method"
          }
        }
      },
      "additionalFields": {
        "type": "MANUAL",
        "schema": {
          "actionPairs": {
            "displayName": "Add Field",
            "required": false,
            "type": "ARRAY",
            "properties": {
              "field": {
                "displayName": "Field",
                "required": true,
                "type": "STATIC_DROPDOWN",
                "options": {"options": []}
              },
              "value": {
                "displayName": "Value",
                "required": true,
                "type": "LONG_TEXT"
              }
            }
          }
        }
      }
    }
  }
}

```

```

    }
  }
}
},
"contentJson": { "type": "MANUAL", "schema": {} },
"identifier": { "type": "MANUAL", "schema": {} },
"identifierNested": { "type": "MANUAL", "schema": {} },
"recordId": { "type": "MANUAL", "schema": {} },
"outputData": { "type": "MANUAL", "schema": {} },
"testing_organization_id": {
  "type": "MANUAL",
  "schema": {
    "testing_organization_id": { "type": "SHORT_TEXT", "required": false, "displayName": "Organization ID" }
  }
},
"errorHandlingOptions": {
  "continueOnFailure": { "value": false },
  "retryOnFailure": { "value": false }
}
}
}

```

WRONG additionalFields format (flat object — causes Cannot read properties of undefined (reading 'length')):

```

"additionalFields": {
  "name": "{{step_1['success'] [0] ['data'] ['name']}}",
  "email": "{{step_1['success'] [0] ['data'] ['email']}}"
}

```

CORRECT additionalFields format (nested with actionPairs array):

```

"additionalFields": { "actionPairs": [
  { "field": "69ca065d5817f89ead6fadf4", "value": "{{step_1['success'] [0] ['data'] ['name']}}",
  { "field": "69ca065d5817f89ead6fadf5", "value": "{{step_1['success'] [0] ['data'] ['email']}}"}
]}

```

ACTION: update — Update Fields

Property	Type	Required
inputMethod	StaticDropdown	YES — MANDATORY
identifier	StaticDropdown	YES — MANDATORY ("recordId" or "fieldsOnBoard")
If identifier = "recordId" → set recordId (nested).		
If identifier = "fieldsOnBoard" → set identifierNested.field + identifierNested.value.		

ACTION: deleteRecord — Delete a Record

Property	Type	Required
identifier	StaticDropdown	YES
Same identifier setup as update.		

ACTION: clearFields — Clear Field Values

Property	Type	Required
identifier	StaticDropdown	YES
additionalFields.actionPairs	Array	No — { field: "<field_id>" } (no value).

ACTION: `getAll` — Get All Records

Property	Type	Required
<code>outputData</code>	StaticDropdown	YES — " <code>allData</code> " or " <code>listRecords</code> "

ACTION: `search` — Get Single Record

Property	Type	Required
<code>identifier</code>	StaticDropdown	YES

If `identifier` = "`recordId`" → `recordId` value.
If `identifier` = "`fieldsOnBoard`" → `identifierNested.additionalFields` array.

ACTION: `boardDetails` — Get Board Details

No additional properties. Returns full board schema with field `_id` values.

3. Field Types and Value Formatting

Field Type	Input Format	Example
ShortText	String (max 150 chars)	"John Doe"
LongText	String	"Full description..."
SingleSelection	String (option <code>_id</code>)	"opt_abc123"
MultipleSelection	Comma-separated or array	"opt_1,opt_2"
Number	Number or numeric string	42 or "42"
Date	ISO date string	"2025-03-15"
Time / Datetime	ISO datetime string	"2025-03-15T14:30:00Z"
Link	URL string	"https://example.com"
Email	Email string	"john@example.com"
Phone	Phone with country code	"+66812345678"
Country	2-letter code or name	"TH" or "Thailand"
Notes	Text string	"Follow up needed"
Attachment	Comma-separated URLs	"https://example.com/file.pdf"
Currency	Number + <code>currencyCode</code> param	Value: 100.50, <code>currencyCode</code> : "USD"
Checkbox	" <code>is_checked</code> " or other	Only " <code>is_checked</code> " = true

4. Property Visibility Decision Tree


```
ALWAYS VISIBLE: workflowName, version, boardId, chooseAction
```

```
chooseAction = "create"
+-- inputMethod (field | json)          <-- MANDATORY, NESTED
+-- IF json --> contentJson              <-- NESTED, FIELD ID keys
+-- IF field --> additionalFields.actionPairs [{field: "<field_id>", value}]

chooseAction = "update"
+-- inputMethod (field | json)          <-- MANDATORY, NESTED
+-- identifier (recordId | fieldsOnBoard) <-- MANDATORY, NESTED
+-- IF json --> contentJson              <-- NESTED, FIELD ID keys
+-- IF field --> additionalFields.actionPairs
+-- IF recordId --> recordId              <-- NESTED
+-- IF fieldsOnBoard --> identifierNested.field + identifierNested.value

chooseAction = "deleteRecord"
+-- identifier                          <-- MANDATORY, NESTED

chooseAction = "clearFields"
+-- identifier                          <-- MANDATORY, NESTED
+-- additionalFields.actionPairs [{field: "<field_id>"}] (no value)

chooseAction = "getAll"
+-- outputData (allData | listRecords) <-- MANDATORY, NESTED

chooseAction = "search"
+-- identifier                          <-- MANDATORY, NESTED

chooseAction = "boardDetails"
+-- (no additional properties)
```

5. Output Schemas by Action

create / update / deleteRecord / clearFields

```
{
  "boardResponse": {
    "_id": "record_id",
    "fields": { "field_id": "value" },
    "public_id": "PUB-001"
  }
}
```

getAll with outputData = "allData"

```
{
  "boardResponse": {
    "data": [
      { "_id": "rec_1", "fields": { "field_id_1": "value1" }, "public_id": "PUB-001" }
    ]
  }
}
```

getAll with outputData = "listRecords" and version = "2"

```
{
  "data": [
    {
      "Field Name 1": { "value": "extracted_value", "type": "ShortText", "_id": "field_id" },
      "_id": "rec_1", "public_id": "PUB-001"
    }
  ]
}
```

search (fieldsOnBoard)

```
{ "boardResponse": [ { "_id": "rec_1", "fields": { "field_id": "value" } } ] }
```

boardDetails

```
{
  "boardResponse": {
    "_id": "board_id", "name": "My Board",
    "fields": [
      { "_id": "field_id", "name": "Status", "type": "SingleSelection", "is_identifier": false,
        "data": [ { "_id": "opt_1", "value": "Active" } ] }
    ]
  }
}
```

6. Common Errors and Troubleshooting

Symptom	Root Cause	Fix
"Identifier value is required"	Used field names instead of field IDs, OR identifier field has empty value	Use 24-char hex field IDs. Ensure identifier field has non-empty value (use <code>trigger.body['file_url']</code>).
inputMethod saved as null	Used flat format	Use nested: <code>{"inputMethod": "json"}</code>
contentJson saved as {}	Missing schema in propertySettings	Include <code>schema</code> block for contentJson
Step in invalidSteps	Missing DYNAMIC nesting or missing propertySettings schema	Refer to FULL CONFIGURATION EXAMPLE
Field value contains [object Object]	Passed object to string field	Reference specific sub-field: <code>step_1['success'][0]['data']['full_name']</code>
"failedsuccess" in field	Used <code>{{#if}}</code> conditional	Use Code step for conditional logic